

Newton’ s cell action as a measure on cuts

After refereeing (Fable, 2026-09-28). Theorems 1–2, Corollary 3’ s arithmetic and the quotations were re-derived and accepted. Corrections applied: Proposition 4 now names the Kullback–Leibler divergence, the shift δx , the phase K_τ/ε and the Cameron–Martin normalizer, and its link to Rivero 1998 is stated as a counterpart; Corollary 5 separates a floor on a cell’ s action from a floor on a cut’ s share (a factor $\frac{3}{4}$ for halving); § 3 labels the two-quarter-cut schedule a construction and adds the alternating reading favoured by 斲必半; the Lévy–Ciesielski comparison names the right quantity.

Result, 2026-09-27 (atlas § 1b, user remarks on arbitrary cuts and on the rod). Let a constant force F act on a body of mass m over a cell of duration τ , and let δx be the chord minus the parabola (the chord is the inertial comparison path with the same endpoints). Its Galileo action

$$K_\tau = \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{F^2 \tau^3}{24m}$$

(the two-path note writes the same quantity as $(F/4v)A = \tau \Delta E/12$) behaves as a measure on the process of cutting the cell.

- **Theorem 1 (one cut).** A cut at fraction $s \in (0, 1)$ spends exactly $3s(1-s)K_\tau$, and this share equals the Cameron–Martin energy $\frac{m}{2} \int \dot{\phi}^2$ of the Schauder hat ϕ that the cut inserts, of height the sagitta $Fs(1-s)\tau^2/(2m)$.
- **Theorem 2 (any cut sequence).** The hats of successive cuts are orthogonal in the energy $\int \dot{f} \dot{g}$, so the spent shares add. Along any sequence of cuts, at any positions and in any order, the total spent equals $K_\tau - \sum_i K_{\tau_i}$ over the current pieces, and it tends to K_τ iff the mesh $\max_i \tau_i$ tends to zero; the remainder is at most $K_\tau \max_i \tau_i^2/\tau^2$.
- **Corollary 3 (the rod and the Mohist schedules).** Halving only the remaining piece each day, the schedule of the stick in Zhuangzi 33, spends $\frac{6}{7}K_\tau$, and so does one halving a day taken alternately from the front and the back. Removing a quarter from each end each day (two quarter-cuts a day, a construction motivated by the Mohist “taking from front and back”) spends $\frac{27}{28}K_\tau$. All three leave a definite share unspent forever, because they keep cut-off pieces of fixed length.
- **Proposition 4 (weight of a cut).** Under the Euclidean path measure of a free particle with diffusion constant $\hbar/m$ (Wiener measure), the force signal in one cut has Kullback–Leibler divergence exactly (spent share)/ $\hbar$. In real time the same share, divided by the action resolution, is the phase of the two-path comparison.
- **Proposition 6 (the mark mesh as a Cameron–Martin threshold; 2026-09-28, not refereed).** Under the same path measure, the optimal equal-prior test between force F and no force in a cell, observing the whole path with pinned ends, errs with probability $\Phi(-\sqrt{K_\tau/2\hbar})$. It reaches ϵ iff $K_\tau \geq 2z_{1-\epsilon}^2 \hbar$, which is exactly the Planck paper’ s mark mesh $\tau_* = (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$, obtained there from real-time Gaussian marks with Robertson’ s bound.
- **Corollary 5 (a floor counts the cuts).** If every exhibited cut must spend at least an action $\kappa > 0$, at most K_τ/κ cuts can be exhibited, whatever the schedule. For halving refinement, level n is exhibited iff $\frac{3}{4}K_\tau 8^{-n} \geq \kappa$. The Planck paper’ s mark mesh $\tau_* = (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$ is the cell whose action is $2z_{1-\epsilon}^2 \hbar$; its own halving spends $\frac{3}{2}z_{1-\epsilon}^2 \hbar$.

Theorems 1 and 2 are elementary, and their content lies in the reading they give: Newton’ s refinement is a process that spends a fixed, finite action, with the nested additivity of the Lévy–Ciesielski construction of the Brownian bridge. Proposition 4

makes the correspondence exact: the action is the Cameron–Martin energy of the force’ s shift, decomposed over the hats. Corollary 5 is a consistency statement. It says where a floor, if present, stops the cutting, and leaves the necessity question of the fifth-postulate note where it was.

1. One cut

Put $C = F^2/(24m)$, so $K_\tau = C\tau^3$. On $[0, \tau]$ the parabola with acceleration F/m and the chord through its endpoints differ by

$$\delta x(t) = \frac{F}{2m} t(\tau - t), \quad \dot{\delta x}(t) = \frac{F}{m} \left(\frac{\tau}{2} - t \right),$$

$$\text{so } \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{m}{2} \cdot \frac{F^2}{m^2} \cdot \frac{\tau^3}{12} = C\tau^3.$$

Cut at $t_c = s\tau$. The polygon through the three points $0, s\tau, \tau$ differs from the chord by the hat ϕ of height $h = \delta x(s\tau) = Fs(1-s)\tau^2/(2m)$, linear on $[0, s\tau]$ and on $[s\tau, \tau]$. The polygon differs from the parabola, on each piece, by the chord-minus- parabola of that piece: $\delta x = \phi + \delta x_1 + \delta x_2$ with δx_j supported on piece j and vanishing at its ends.

Orthogonality. $\dot{\phi}$ is constant on each piece and $\int_{\text{piece}} \dot{\delta x}_j = 0$, so $\int \dot{\phi} \dot{\delta x}_j = 0$; the δx_j have disjoint supports. Hence

$$K_\tau = \frac{m}{2} \int \dot{\phi}^2 + K_{s\tau} + K_{(1-s)\tau}.$$

The hat’ s energy. $\frac{m}{2} \int \dot{\phi}^2 = \frac{m}{2} h^2 \left(\frac{1}{s\tau} + \frac{1}{(1-s)\tau} \right) = \frac{mh^2}{2s(1-s)\tau} = \frac{F^2 s(1-s)\tau^3}{8m} = 3s(1-s)C\tau^3$. The same number follows from $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, which is the algebraic form of Theorem 1. The share is largest for halving, $\frac{3}{4}K_\tau$. $\square$

2. Any cut sequence

Each later cut acts inside one current piece, and Theorem 1 applies to that piece with its own s . The new hat is supported in the piece and vanishes at its ends; every earlier hat is linear there. The argument of §1 gives orthogonality to all earlier hats and to the other pieces’ remainders. After any finite set of cuts, with pieces τ_i ($\sum \tau_i = \tau$),

$$K_\tau = \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2 + \sum_i K_{\tau_i}, \quad \sum_i K_{\tau_i} = C \sum_i \tau_i^3 \leq C\tau \max_i \tau_i^2.$$

The remainder vanishes iff the mesh does ($\sum_i \tau_i^3 \geq (\max_i \tau_i)^3$). The positions and the order enter only through which pieces are cut, never through the total. $\square$

This is the Schauder (Lévy–Ciesielski) expansion of δx in the Cameron–Martin space $H_0^1[0, \tau]$, adapted to an arbitrary nested cut sequence; for dyadic cuts it is the textbook expansion (Ciesielski 1961; metadata). The reachability note has the same cubic law for the area $2F^2t^3/(3m)$ of the reachable lens of a cut. The bridge’ s own conditional variance shows the same nested additivity with a quadratic law: for diffusion constant D , $\int_0^L \text{Var}(X_u \mid \text{ends}) du = DL^2/6$, and a cut at fraction s removes $Ds(1-s)L^2/3$ of it.

3. The rod and the two Mohist schedules

Zhuangzi 33 (dialecticians’ list, text): 一尺之捶，日取其半，萬世不竭， “a one-foot stick, each day take half, in ten thousand generations it is not exhausted.” Mohist Canon B (B60 in Graham’ s numbering; canon): 非半弗斷，則不動，說在端， “without halving there is no cutting and no moving; the explanation lies in the point (端)” ,

with the explanation (text) 前則中無爲半，猶端也。前後取，則端中也 (working gloss: “going forward, at the middle nothing serves as a half: it is like a point; taking from front and back, the point is in the middle”), which continues 斲必半, “cutting must be by halves”.

One-ended schedule. Day k halves the remaining piece, of length $\tau 2^{-(k-1)}$, and spends $\frac{3}{4}C\tau^3 8^{-(k-1)}$. The total is $\frac{3}{4} \cdot \frac{8}{7}K_\tau = \frac{6}{7}K_\tau$; the pieces taken away keep $\sum_{k \geq 1} 8^{-k}K_\tau = \frac{1}{7}K_\tau$.

Alternating schedule (the reading favoured by 斲必半). One halving a day, taking the half alternately from the front and from the back. Each day is one halving of the remaining piece, so the spending is that of the one-ended schedule, $\frac{6}{7}K_\tau$; the remainder converges to the point $\frac{\tau}{2} + \frac{\tau}{8} + \dots = \frac{2\tau}{3}$.

Two quarter-cuts a day (a construction motivated by 前後取; the text does not fix the fractions). Day k cuts the remaining piece $r = \tau 2^{-(k-1)}$ at $s = \frac{1}{4}$ and $s = \frac{3}{4}$, keeping the middle half. The two cuts spend $C\tau^3(1 - 2 \cdot \frac{1}{64} - \frac{1}{8}) = \frac{27}{32}C\tau^3$ (equivalently $\frac{9}{16} + \frac{9}{32}$ by two applications of Theorem 1), and the total is $\frac{27}{32} \cdot \frac{8}{7}K_\tau = \frac{27}{28}K_\tau$, with $\frac{1}{28}K_\tau$ kept in the pieces taken away. The contrast $\frac{6}{7}$ against $\frac{27}{28}$ is between one half-cut and two quarter-cuts a day, whichever ends are cut.

All three schedules converge to a point, the Mohist 端: the end, the point $2\tau/3$, and the midpoint. If 中 in 端中 means the midpoint, the explanation fits the symmetric schedule, while 斲必半 fits the alternating one. What each author is committed to, and what the theorem does with it:

- The dialecticians hold that the halving never ends. Theorem 2 agrees for the figure and adds a price: their schedule spends only $\frac{6}{7}$ of the action, since it never cuts again what it took away.
- The Mohists hold that the halving ends at a point. Theorem 2 agrees that each schedule converges to a point, and the point carries no action. The unhalvable 端 as a least magnitude is a reading the theorem leaves aside, as the Planck paper already records.
- Newton’s scholium closing Book I, Section I, takes the geometers’ side (divisibility without end). Theorem 2 is that side made quantitative: refinement exhausts the action only when every piece is cut again.

The parallel with Zeno’s dichotomy (Aristotle, Physics VI.9) is recorded as convergence; transmission would need evidence this note does not have.

4. The weight of a cut

Take the Euclidean path measure of a free particle with diffusion constant $\hbar/m$: Brownian motion X with $E[X_t^2] = \hbar t/m$, pinned at the cell’s endpoints. Given the endpoints of a piece of length L , the value at fraction s is Gaussian with variance $\hbar s(1-s)L/m$, independent of the finer hats and of the other pieces, by the Markov property of the bridge (the Lévy construction, for any nested sequence). A force F shifts the mean of that value by the sagitta $h = Fs(1-s)L^2/(2m)$. The Kullback–Leibler divergence of the shifted from the free law (the expected log-likelihood ratio, which here also equals the ratio evaluated at the shift) is

$$\frac{h^2}{2\hbar s(1-s)L/m} = \frac{mh^2}{2s(1-s)L\hbar} = \frac{3s(1-s)CL^3}{\hbar},$$

the spent share of Theorem 1 divided by $\hbar$. By the chain rule the divergences add over cuts, and along a sequence with mesh tending to zero they sum to $K_\tau/\hbar$, the Cameron–Martin exponent for the shift δx . The Cameron–Martin density factorizes

over cuts, each contributing the normalizing factor $e^{-3s(1-s)CL^3/\hbar}$. This is the Euclidean counterpart of the real-time two-path phase K_τ/ε of the two-path note (the same number with ε in the role of $\hbar$), and of the product-over-cuts structure that § 7 of the fifth-postulate note reads in the time-step partition of Rivero 1998. For $\hbar \rightarrow 0$ every cut, however small, is decisive; for $\hbar > 0$ the cuts whose share falls below $\hbar$ carry little evidence (under one nat). $\square$

5. A floor counts the cuts

If every cut that a record exhibits must spend at least $\kappa > 0$, Theorem 2 gives at most K_τ/κ such cuts, for every schedule. For halving refinement the 2^n cuts of level n each spend $\frac{3}{4}K_\tau 8^{-n}$, so the exhibited levels are those with $8^n \leq 3K_\tau/(4\kappa)$.

Two floors must be kept apart: κ here floors the share a cut spends, $3s(1-s)K_L$, while the thresholds in the other notes floor the action of a cell, K_τ ; for halving the two differ by the factor $\frac{3}{4}$. The Planck paper's eq. (2) is $K_\tau \geq 2z_{1-\epsilon}^2\hbar$: τ_* is the cell with $K_{\tau_*} = 2z_{1-\epsilon}^2\hbar$, whose halving spends $\frac{3}{2}z_{1-\epsilon}^2\hbar$. The round-3 resolution threshold $\tau \geq (24m\varepsilon d_{n,\eta}/F^2)^{1/3}$ is $K_\tau \geq \varepsilon d_{n,\eta}$, that is $\kappa = \frac{3}{4}\varepsilon d_{n,\eta}$ in cut-share form. Corollary 5 thus places a floor, when one is given, at a definite depth of the cut process; it supplies no floor.

5b. The mark mesh from the path measure (Proposition 6)

Keep the measure of § 4: Brownian motion with $E[X_t^2] = \hbar t/m$, the Euclidean free-particle measure, whose diffusion coefficient $\hbar/2m$ is the one of Nelson (1966) (meta-data). For the whole cell, the laws with and without the force are Gaussian measures differing by the shift δx , with Cameron–Martin norm $\|\delta x\|^2 = (m/\hbar) \int \dot{\delta x}^2 = 2K_\tau/\hbar$. The log-likelihood ratio is the affine function $\langle \delta x, X \rangle - \frac{1}{2}\|\delta x\|^2$ of one Gaussian statistic of variance $\|\delta x\|^2$ whose mean moves by $\|\delta x\|^2$, so the Neyman–Pearson test with equal priors thresholds that statistic halfway and errs with probability

$$p_{\text{err}} = \Phi\left(-\frac{1}{2}\|\delta x\|\right) = \Phi\left(-\sqrt{K_\tau/2\hbar}\right).$$

Hence $p_{\text{err}} \leq \epsilon$ iff $K_\tau \geq 2z_{1-\epsilon}^2\hbar$, that is $\tau\Delta E = 12K_\tau \geq 24z_{1-\epsilon}^2\hbar$ and $\tau \geq \tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$: eq. (2) of the Planck paper, including its “about $65\hbar$ at $\epsilon = 0.05$ ” ($2z_{0.95}^2 = 5.41$). A single cut observed alone gives $\Phi(-\sqrt{\text{share}/2\hbar})$ in the same way. $\square$

The Planck paper reaches (2) from real-time Gaussian marks whose record and recoil obey Robertson's bound $\hbar/2$; here the same constant comes from the Euclidean path measure $e^{-S/\hbar}$ through the Cameron–Martin theorem alone. The agreement is exact, and both routes supply $\hbar$: this is a second derivation of the mesh, with the necessity question of the fifth-postulate note unchanged. What it adds is the reading of the floor on recording the inertial–parabola difference as the distinguishability threshold of the force in the free-particle path measure. At small ϵ , $z_{1-\epsilon}^2 \simeq 2\ln(1/\epsilon)$, so this floor grows like $4\hbar \ln(1/\epsilon)$ in cell action, the same logarithmic law as Theorem E of the fifth-postulate note, $ab \geq \hbar c_*(\epsilon)$ with $c_*(\epsilon) \simeq \frac{1}{2}\ln(1/\epsilon)$ in window half-widths; the constants refer to different quantities (a cell's Galileo action and a phase-space window).

6. Consequence for STATE

Atlas § 1b: the Newton row of “cuts at any position” is proved here (Theorems 1–2, Corollary 3), with the Lévy–Ciesielski identification and the Euclidean weight of a cut (Proposition 4). The necessity question is unchanged. On the gauge side of § 1b the

free-field defect of a cut at fraction s is (5) scaled by $4s(1-s)$ in its size bound (Corollary 2_s); the $U(1)$ Theorem 5 holds for every cut fraction (Corollary 5_s there).